

SIMATS ENGINEERING
CO2_ASSESSMENT TOOL3_Application Based Questions

COURSE NAME: Cloud Computing and Big Data Analytics **COURSE CODE:** CSA1522

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COURSE OUTCOME COVERED

CO2: Analyze virtualization architectures to identify performance and security issues in cloud computing environments. (BL3)

Dave's Taxonomy: Manipulation (Level 2)
Question Count: 5

Total Marks: 25

Code of Conduct

I **M. Meghana(192525435)** certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: *M. Meghana*

Assessment Tasks

Application Based Questions

1. A company experiences unpredictable traffic spikes during seasonal sales. Explain how cloud auto-scaling and load balancing can handle this demand efficiently. How does this approach reduce downtime and improve user experience? What role do monitoring tools play in this setup? Relate your answer to a real-world e-commerce scenario.
2. An organization wants to store large volumes of backup data securely in the cloud. Discuss how cloud storage solutions ensure durability and data redundancy. How can encryption and access control protect sensitive backup data? Explain the cost advantages compared to traditional storage systems. Provide an example of a disaster recovery use case.
3. A mobile app company plans to deploy its application globally. Explain how content delivery networks (CDNs) improve performance and reduce latency. How do edge

servers enhance user experience across regions? Discuss how cloud providers support global availability. Relate this to a real-time streaming or gaming application.

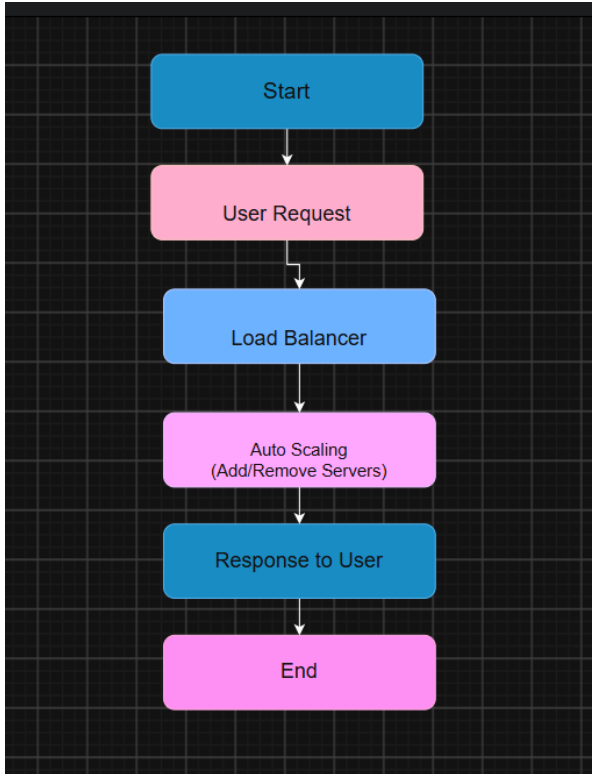
4. A financial institution requires strict compliance and data security in cloud usage. Explain how cloud providers support regulatory compliance and auditing. What role do logging, monitoring, and encryption play in security? How can the company ensure data sovereignty requirements are met? Give an example involving sensitive financial transactions.
5. A development team wants to adopt containerization for application deployment. Explain how containers differ from virtual machines in terms of performance and resource usage. How does orchestration (like Kubernetes) simplify deployment and scaling? Discuss portability benefits across different cloud environments. Provide an example of a microservices-based application.

Application Based Questions Rubric

Criteria	Weightage	Level 4 - Exemplary	Level 3 - Proficient	Level 2 - Developing	Level 1 - Beginning
Relevance to Application	25%	Response directly addresses the given use case with strong relevance	Mostly relevant response	Partially relevant	Weak relevance
Use of Cloud Concepts	25%	Applies appropriate concepts accurately	Mostly accurate application	Basic application only	Incorrect concept usage
Practical Insight	20%	Shows strong real-world understanding	Shows reasonable practical understanding	Limited practical insight	No practical insight
Completeness	15%	Response covers all required aspects	Covers most aspects	Covers only basic aspects	Incomplete response
Clarity	15%	Clear and concise answer	Generally clear	Some confusion in expression	Poor clarity

1. A company experiences unpredictable traffic spikes during seasonal sales. Explain how cloud auto-scaling and load balancing can handle this demand efficiently. How does this approach reduce downtime and improve user experience? What role do monitoring tools play in this setup? Relate your answer to a real-world e-commerce scenario.

Cloud Auto-Scaling and Load Balancing for Seasonal Sales



During seasonal sales such as Black Friday or Diwali, e-commerce websites receive sudden increases in user traffic. Cloud computing uses **auto-scaling** and **load balancing** to manage this demand efficiently.

Auto-Scaling

- Automatically increases or decreases the number of servers based on traffic.
- Adds new virtual machines when demand is high.
- Removes extra resources when traffic decreases.
- Ensures applications remain responsive.

Load Balancing

- Distributes incoming requests evenly across multiple servers.
- Prevents any single server from becoming overloaded.
- Improves application availability and reliability.

- Ensures smooth user access.

Reducing Downtime

- Prevents server crashes during heavy traffic.
- Reduces website loading time.
- Provides uninterrupted shopping experience.
- Increases customer satisfaction and sales.

Role of Monitoring Tools

- Continuously monitor CPU, memory, and network usage.
- Detect performance issues in real time.
- Trigger auto-scaling when resource usage exceeds thresholds.
- Send alerts for failures or abnormal activity.

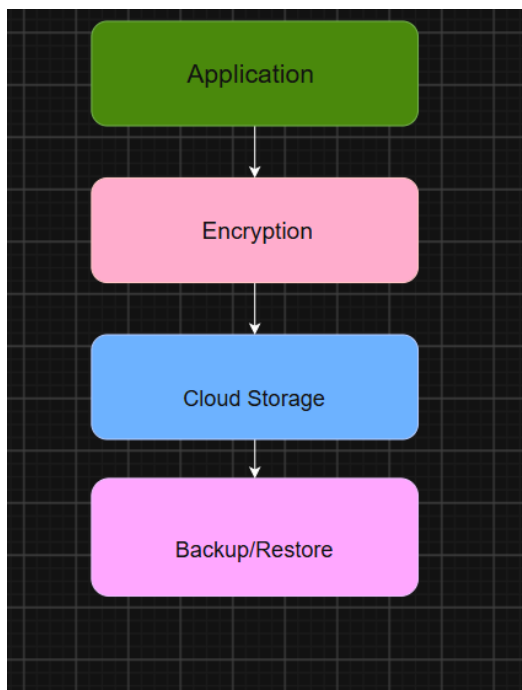
Real-World Example

During **Amazon Prime Day**, millions of customers shop simultaneously. Auto-scaling creates additional servers automatically, while load balancers distribute customer requests, ensuring fast and reliable service.

2. An organization wants to store large volumes of backup data securely in the cloud. Discuss how cloud storage solutions ensure durability and data redundancy. How can encryption and access control protect sensitive backup data? Explain the cost advantages compared to traditional storage systems. Provide an example of a disaster recovery use case.

Cloud Storage for Secure Backup Data

Organizations store large amounts of backup data in cloud storage because it is secure, reliable, and cost-effective.



Durability and Data Redundancy

- Data is stored across multiple storage devices.
- Multiple copies are maintained in different locations.
- Hardware failures do not cause data loss.
- Automatic replication ensures high availability.

Access Control

- Encrypts data during storage and transmission.
- Only authorized users can access backup files.
- Multi-factor authentication (MFA) improves security.
- Identity and Access Management (IAM) controls permissions.

Advantages

- No need to purchase expensive storage hardware.
- Pay only for the storage used.
- Reduced maintenance costs.
- Easy scalability without infrastructure investment.

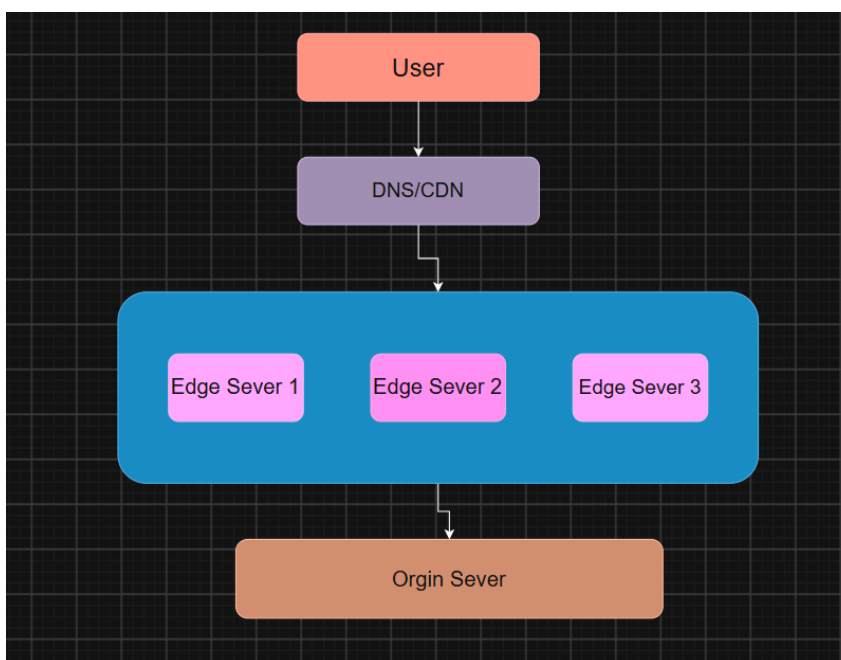
Example:

A company experiences a ransomware attack that encrypts local files. Cloud backups allow the organization to restore clean data quickly, minimizing downtime and business loss.

3.A mobile app company plans to deploy its application globally. Explain how content delivery networks (CDNs) improve performance and reduce latency. How do edge servers enhance user experience across regions? Discuss how cloud providers support global availability. Relate this to a real-time streaming or gaming application.

Global Application Deployment Using CDN

A **Content Delivery Network (CDN)** improves application performance by delivering content from servers located close to users.



Benefits of CDN

- Reduces network latency.
- Faster loading of web pages and media.
- Decreases bandwidth usage.
- Handles high traffic efficiently.

Role of Edge Servers

- Store cached copies of frequently accessed content.
- Deliver content from the nearest geographical location.
- Reduce response time.
- Improve user experience worldwide.

Global Availability

Cloud providers maintain data centre across multiple regions.

- Automatic routing to nearest server.
- Failover during regional outages.
- High availability.
- Better disaster recovery.

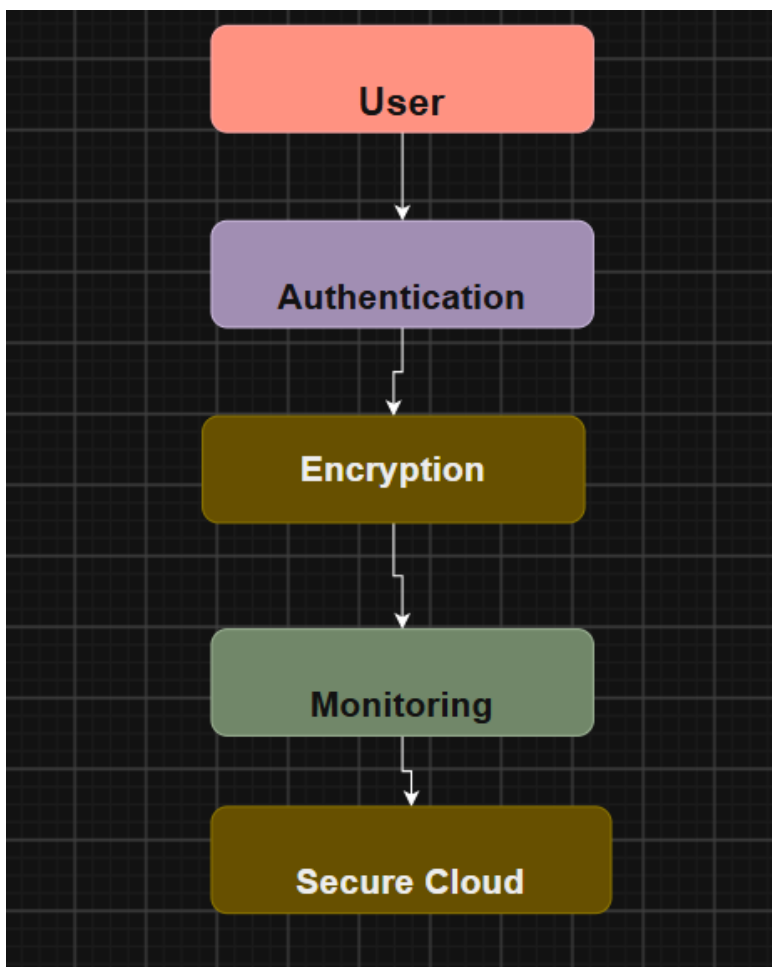
Real-Time Streaming

A video streaming platform like Netflix delivers movies through CDN edge servers. Users in different countries receive content from nearby servers, reducing buffering and improving streaming quality.

4.A financial institution requires strict compliance and data security in cloud usage. Explain how cloud providers support regulatory compliance and auditing. What role do logging, monitoring, and encryption play in security? How can the company ensure data sovereignty requirements are met? Give an example involving sensitive financial transactions

Cloud Security and Regulatory Compliance

Financial institutions require strong security and compliance because they handle sensitive customer and transaction data.



Regulatory Compliance

Cloud providers support standards such as:

- ISO 27001
- PCI DSS
- GDPR

- SOC 2

These certifications help organizations meet legal and industry requirements.

Logging

- Records all user activities.
- Supports auditing.
- Detects unauthorized access.

Monitoring

- Identifies suspicious behavior.
- Generates security alerts.
- Supports incident response.

Encryption

- Protects stored and transmitted data.
- Prevents unauthorized data access.
- Uses secure encryption algorithms.

Data Sovereignty

- Stores customer data within required countries.
- Allows organizations to select cloud regions.
- Meets government regulations regarding data location.

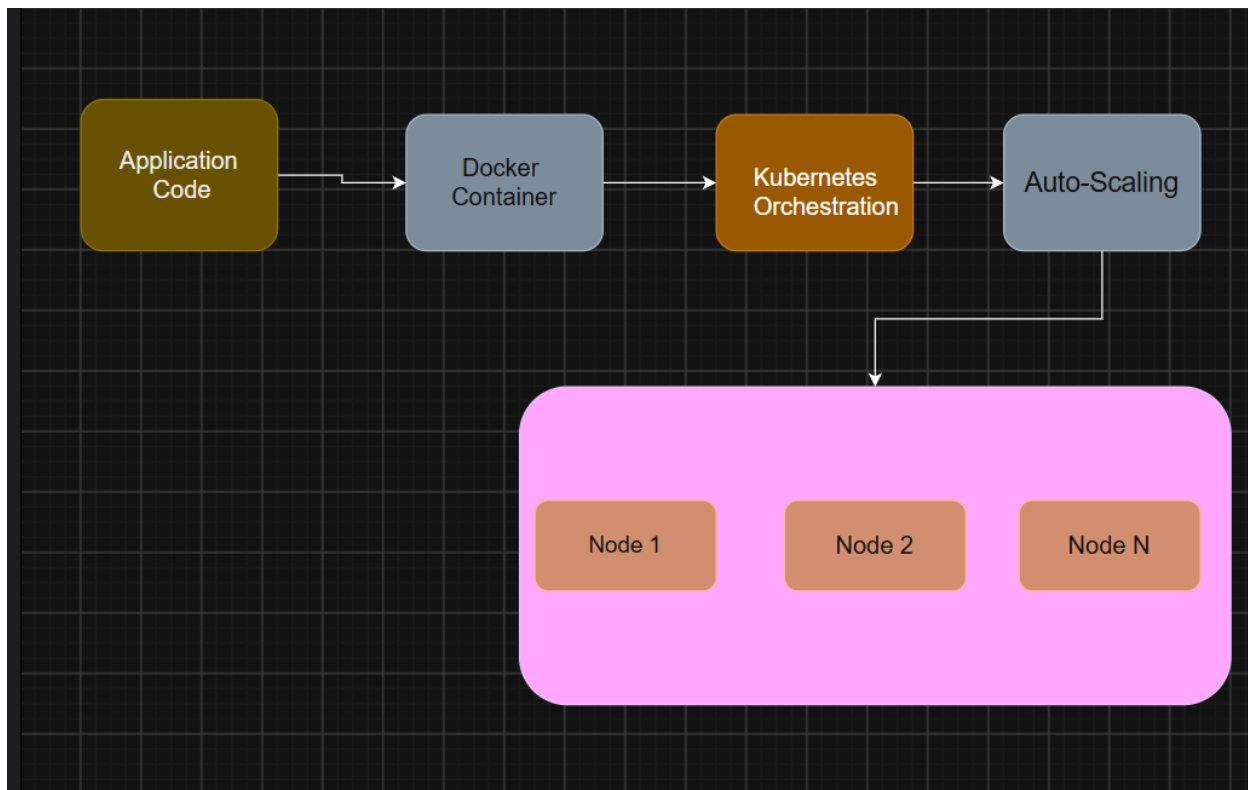
Financial Transaction Example

An online banking system encrypts customer transactions, stores audit logs, and continuously monitors suspicious activities while keeping financial records within the country's approved data centers to comply with regulations.

5.A development team wants to adopt containerization for application deployment. Explain how containers differ from virtual machines in terms of performance and resource usage. How does orchestration (like Kubernetes) simplify deployment and scaling? Discuss portability benefits across different cloud environments. Provide an example of a microservices-based application.

Containerization and Kubernetes

Containerization packages applications with all their dependencies, making deployment faster and more consistent.



Containers	Virtual Machines
Lightweight	Heavyweight
Share host operating system	Each VM has its own OS
Fast startup	Slower startup
Lower resource usage	Higher resource usage
Better performance	More overhead

Kubernetes Orchestration

Kubernetes automates:

- Container deployment.
- Load balancing.
- Auto-scaling.
- Self-healing by restarting failed containers.
- Rolling updates with minimal downtime.

Benefits

- Applications run consistently across different cloud providers.
- Easy migration between AWS, Azure, and Google Cloud.
- Simplifies hybrid and multi-cloud deployments.

Example

An online shopping application can be divided into:

- User Service
- Product Catalog Service
- Shopping Cart Service
- Payment Service
- Order Service
- Notification Service

Each service runs in its own container and Kubernetes manages deployment, scaling, and availability independently.